review

June 11, 2025

[3]:
```
import pandas as pd
import matplotlib.pyplot as plt

# convert the file into a csv file
df = pd.read_excel('DeepLearing_Evapotranspiration_Papers.xlsx')
df.to_csv('review.csv', index=False)
```

[4]:
```
# read the csv file
df = pd.read_csv('review.csv')

# print the column names
print(df.columns)
```

```
Index(['Publication Type', 'Authors', 'Book Authors', 'Book Editors',
       'Book Group Authors', 'Author Full Names', 'Article Title',
       'Source Title', 'Language', 'Document Type',
       …
       'Linear detrending', 'RMSprop', 'SGD', 'Adam', 'short term', 'mid term',
       'long term', 'Local', 'Regional', 'Global.1'],
      dtype='object', length=307)
```

[5]:
```
# # remove all the columns before the column 'DNN'
df = df.loc[:, 'DNN':]

# convert the values in the dataframe to boolean
df = df.applymap(lambda x: True if x == 'True' else True if x == True else␣
 ↪False)

# write the dataframe to a new csv file
df.to_csv('review.csv', index=False)
```

```
/tmp/ipykernel_52222/3824094904.py:5: FutureWarning: DataFrame.applymap has been
deprecated. Use DataFrame.map instead.
  df = df.applymap(lambda x: True if x == 'True' else True if x == True else
False)
```

[6]:
```
algorithms = [
    'DNN', 'MLP', 'DeepFM', 'BPNN', 'GRNN', 'DBN', 'DL', 'RNN', 'BiRNN',␣
 ↪'LSTM', 'BiLSTM',
```

```
    'GRU', 'BiGRU', 'SSR-LSTM', 'ConvLSTM', 'NARX', 'CNN', 'YOLO',␣
 ↪'Transformer', 'ADAQNet',
    'QAxNet', 'GAN', 'AE', 'CVAE', 'ELM', 'K-means-FFA-KELM', 'ANFIS', 'MPS',␣
 ↪'QSDM', 'GMDH',
    'SWAT', 'SL', 'PLR', 'eXML', 'MLNR', 'CBR', 'DT', 'CART', 'BCART',␣
 ↪'M5Tree', 'RF', 'DRF',
    'BT', 'GB', 'XGB', 'DF', 'LightGB', 'AdaBoost', 'LR', 'Lasso', 'PL', 'GLM',␣
 ↪'GPR', 'SARIMA',
    'GLM', 'MEMD', 'MARS', 'SAR2ET', 'HBV', 'BMA', 'PT-JPL', 'SR', 'ER', 'SVM',␣
 ↪'SVR', 'RL', 'DRL',
    'MTL', 'STL', 'AutoML'
]

metrics = [
    'RMSE', 'MSE', 'RMSD', 'SEP', 'ubRMSE', 'SDR', 'WSEE', 'E_l,m',
    'MAE', 'WAPE', 'MAD', 'MADE', 'AARE', 'MAPE', 'MdAPE', 'BME',
    'Bias', 'MBE', 'RB', 'RRMSE', 'NRMSE', 'RRSE', 'RAE', 'R',
    'R^2', 'VAF', 'PCC', 'KGE', 'NSE', 'NNSE', 'IOA',
    'Accuracy', 'Precision', 'F1 Score', 'AUC', 'IOU'
]

dataSources = [
    'MODIS', 'Sentinel-1', 'Sentinel-2', 'AVHRR', 'NOAA CDR AVHRR NDVI',␣
 ↪'ASTER', 'SMOS', 'AMSR2', 'OCO-2', 'GRACE',
    'HyTES', 'ECOSTRESS', 'COMS', 'SST', 'GLASS', 'HLS', 'SRTM', 'Copernicus',␣
 ↪'ERA 5', 'MERRA2', 'NLDAS', 'GLDAS',
    'FLDAS', 'NCA-LDAS', 'WFDEI', 'ACCESS-ESM1,5', 'CMIP6', 'CHIRPS', 'SPEI',␣
 ↪'GIOVANNI', 'POWER', 'GLEAM', 'LAADS',
    'EEFLUX', 'NESDC', 'AggieAir sUAS', 'ASDC', 'CSIF', 'PlantVillage',␣
 ↪'FLUXNET', 'SNOTEL', 'ISMN', 'CIMIS',
    'AERONET', 'GRDC', 'USGS', 'INMET', 'HPRCC', 'AZMET', 'FAWN', 'DEOS',␣
 ↪'AWN', 'Weather station', 'Lysimeter',
    'Eddy Covariance', 'CAMELS', 'CANOPEX', 'TAMARACK', 'CRC', 'ACARR', 'LULC',␣
 ↪'DEM', 'HydroSHEDS', 'NALCMS', 'ASCE'
]

locations = [
    'Iran', 'Iraq', 'Egypt', 'Canada', 'Australia', 'U.S.', 'China',␣
 ↪'Colombia', 'India', 'South Korea',
    'Pakistan', 'Bangladesh', 'Israel', 'Vietnam', 'Laos', 'Cambodia',␣
 ↪'Indonesia', 'Malaysia', 'Nepal', 'Slovakia',
    'Sudan', 'Nigeria', 'Ethiopia', 'Algeria', 'Italy', 'Germany', 'Turkey',␣
 ↪'Morocco', 'Peru', 'Ecuador', 'Mexico',
    'Chile', 'Brazil', 'South Africa', 'Portugal', 'Sweden', 'Global', 'Aegean␣
 ↪areas', 'Balkans areas', 'Africa',
```

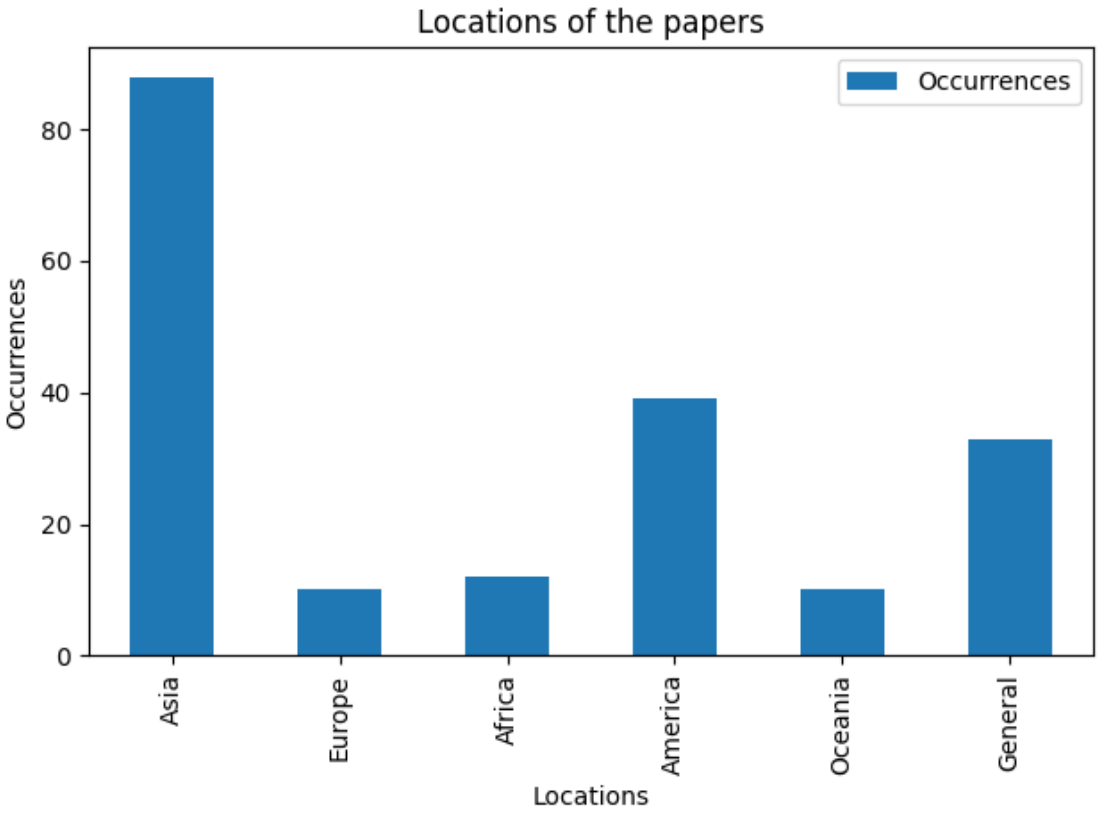
```
    'Amazon', 'Mediterranean areas', 'Oceania areas', 'Savannahs',␣
 ↪'Grasslands', 'Wetlands', 'Croplands', 'Forests'
]

asianLocations = [
    'Iran', 'Iraq', 'China', 'India', 'South Korea', 'Pakistan', 'Bangladesh',␣
 ↪'Israel',
    'Vietnam', 'Laos', 'Cambodia', 'Indonesia', 'Malaysia', 'Nepal', 'Turkey'
]

africanLocations = [
    'Egypt', 'Sudan', 'Nigeria', 'Ethiopia', 'Algeria', 'Morocco', 'South␣
 ↪Africa', 'Savannahs', 'Africa'
]

americanLocations = [
    'Canada', 'Colombia', 'Peru', 'Ecuador', 'Mexico', 'Chile', 'Brazil',␣
 ↪'Amazon', 'U.S.'
]

europeanLocations = [
    'Italy', 'Germany', 'Slovakia', 'Portugal', 'Sweden', 'Aegean areas',
    'Balkans areas', 'Mediterranean areas'
]

oceanianLocations = [
    'Australia', 'Oceania areas'
]

generalLocations = [
    'Grasslands', 'Wetlands', 'Croplands', 'Forests', 'Global'
]

variables = ['Meterological variables', 'Soil variables']

optimizations = [
    'GWO', 'HHO', 'PSO', 'SAO', 'WSO', 'FCM', 'K-means', 'PCA', 'ICA', 'SHAP',␣
 ↪'PACF',
    'Transfer Learning', 'Low-pass', 'High-pass', 'Normalization',␣
 ↪'Interpolation', 'Linear detrending',
    'RMSprop', 'SGD', 'Adam'
]

studyTerms = [
    'short term', 'mid term', 'long term'
]
```

```
coverage = [
    'Local', 'Regional', 'Global'
]
```

[17]:
```
def plot_histogram(df, metrics_of_interest, focused_classes):
    mask = df[metrics_of_interest].any(axis=1)
    df_metrics = df[mask]
    counts = df_metrics[focused_classes].sum()
    # counts = counts / counts.sum()  # Normalize to sum to 1
    asianLocationsCounts = counts[asianLocations].sum()
    europeanLocationsCounts = counts[europeanLocations].sum()
    africanLocationsCounts = counts[africanLocations].sum()
    americanLocationsCounts = counts[americanLocations].sum()
    oceanianLocationsCounts = counts[oceanianLocations].sum()
    generalLocationsCounts = counts[generalLocations].sum()
    # put everything into one dataframe
    combined_counts = pd.DataFrame({
        'Asia': asianLocationsCounts,
        'Europe': europeanLocationsCounts,
        'Africa': africanLocationsCounts,
        'America': americanLocationsCounts,
        'Oceania': oceanianLocationsCounts,
        'General': generalLocationsCounts
    }, index=[0])
    combined_counts = combined_counts.T
    combined_counts.columns = ['Occurrences']
    combined_counts.index.name = 'Locations'
    plt.figure(figsize=(12, 5))
    combined_counts.plot(kind='bar')
    plt.title(f'Locations of the papers')
    plt.xlabel('Locations')
    plt.ylabel('Occurrences')
    plt.xticks(rotation=90)
    plt.tight_layout()
    plt.savefig('locations_histogram.png', dpi=300)
    plt.show()
```

[18]:
```
plot_histogram(df, df.columns, locations)
```

```
<Figure size 1200x500 with 0 Axes>
```


Locations of the papers
Occurrences
80
60
40
20
0
Occurrences
Asia
Europe
Africa
America
Oceania
General
Locations